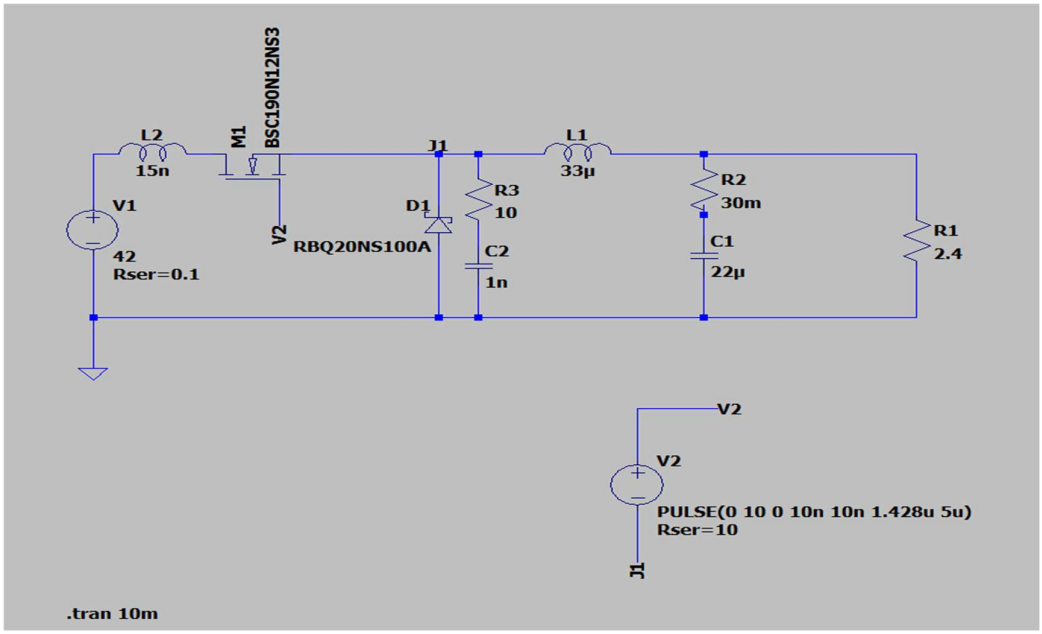
Buck converter design document for non-ideal case (Version 2)

1. **Objective:** Simulate and analyse the non-ideal performance of 60W buck converter in LT Spice by including practical parasitic elements and switching transient mitigation circuitry.

2. **Specifications:**
   - Input voltage(Vin): 42 V
   - Output voltage(Vout): 12 V
   - Output Current(Iout): 5 A
   - Output power(Pout): 60 W
   - Switching frequency(Fsw): 200KHz
   - Load resistance(Rl): 2.4Ω
   - Input source resistance: 100mΩ
   - Input parasitic inductance: 15nH
   - Inductor DCR: 50mΩ
   - Capacitor ESR: 30mΩ
   - Gate resistor: 10Ω
   - Snubbers (RC): 10Ω / 1nF

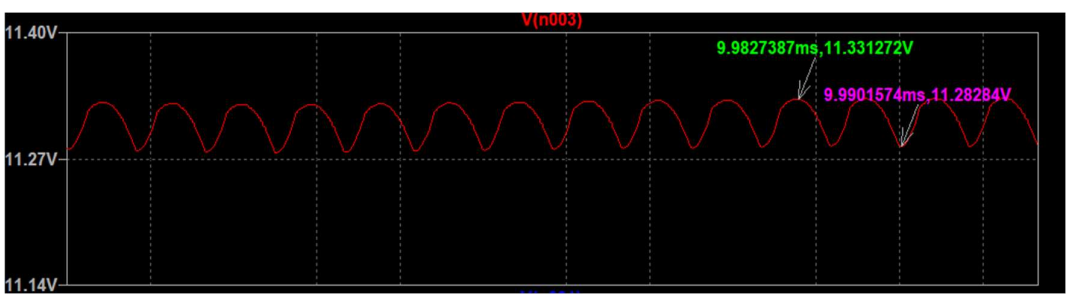
3. **Design calculation (Non-ideal case):**
   - Duty Cycle(D): Vout/Vin: 12/42 = 0.2857
   - Inductor Selected: 33µH: (L(ΔIL)/DT = (V_{in}-V_o), (L = ((42-12) * 0.2857/1.5*200*10^^{3}) = 28.57µH. So, nearest standard inductor available is 33µH.)
   - Estimate Inductor ripple: 1.4A – 1.6A (~30% of Imax)
   - Capacitor selected: 22µF = (ΔV$_o$ = 1/2 * T/2 * ΔIL/2) = (ΔV$_o$ = ΔIL/8fc) : (C = ΔIL/8fΔV$_o$) = (1.5 / (8*200*10^3*0.05) = 18.75µF. So, nearest standard capacitor is 22 µF. Here 0.05V because we want ripple < 50mv.
   - MOSFET selected: BSC190N12NS3(Infineon): (Imax > 8.6A (5.75 * 1.5 (Saf. Margin): 8.6A under typical board-limited thermal condition), (Vds > 42*2(Saf. Margin): 120V), (Rds(on) < 25mΩ to reduce conduction loss: 19mΩ), (Lower figure of merit (Rds(on) * Qg compactible with 200KHz switching frequency).
   - Diode selected: Schottky (RBQ20NS100A – Rohm): (Very low/near-zero reverse recovery ideal for high-frequency switching), (Forward current > 8.6A (5.75*1.5 (Saf. Margin): 20A), (Breakdown voltage > 42*2(Saf. Margin): 100V).

4. Circuit Visual:

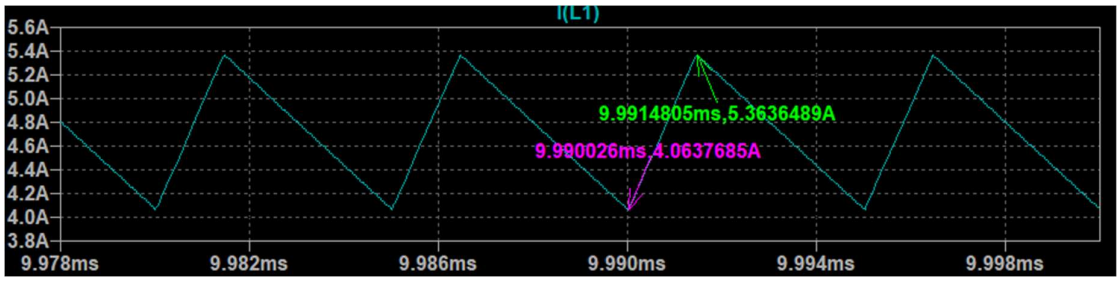


5. Simulation results:

- **Actual output (Vout):** The simulated output voltage varies from 11.3312V to 11.2828V with an average voltage 11.307V. The resulting output voltage ripple is 48.4mv(11.3312-11.2828 = 0.0484V) peak to peak 0.43%((0.0484/11.307)*100), indicating stable regulation with low output voltage ripple under non-ideal operating conditions.

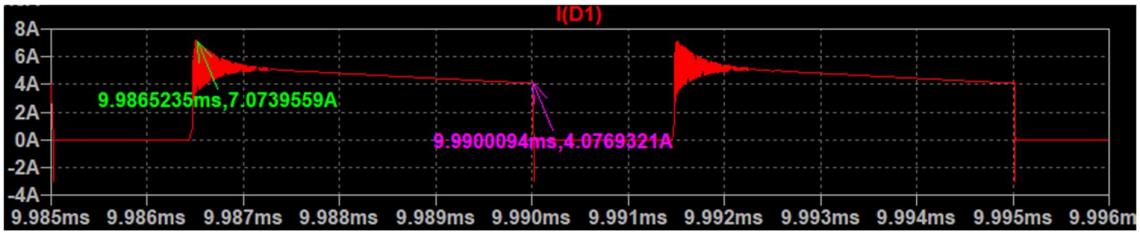


- **Inductor current (IL):** Current varies from 5.3636A to 4.0637A with an average current 4.7137A. The resulting inductor current ripple is 1.2999A(5.3636 – 4.0637 = 1.2999) peak to peak corresponding to 27.6% ((1.2999 / 4.7137) *100) of the average inductor current. This confirms continuous conduction mode (CCM) operation with acceptable ripple current for the selected inductor value.

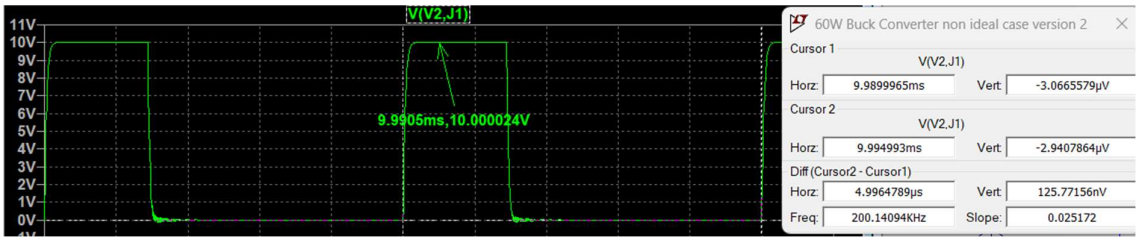


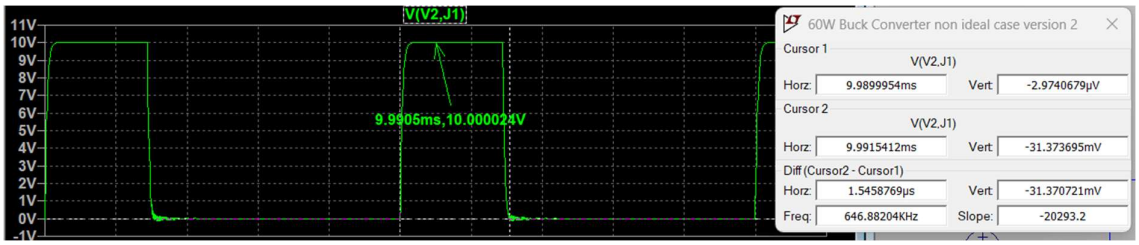
- **Switch node voltage (Vj1**): The Switch-node voltage toggles between approximately 0V to 41.41V, representing the MOSFET switching operation between on and off state.
  A transient overshoot of 49.41V is observed during turn-off corresponding to approx. 7.41V overshoot above input voltage.

- **Diode Current (Id):** The diode conducts during the MOSFET off interval, carrying the inductor current to maintain continuous load current. The diode current varies approx. from 4.07 to 7.07 A.



- **Duty cycle**: The duty cycle of buck converter was measured from gate to source voltage waveform. The switching period(T) is measured as 4.996µs, corresponding to switching frequency 200.14KHz. while the measured-on time is 1.54µs.





6. Result analysis:

- **Inductor current analysis**: The inductor current waveform exhibits a triangular waveform compare to our ideal condition the peak and average current is slightly decreased as a consequence of the Vout drop – parasitics reduce Vout and since Iout = Vout/RL, the load current reduces accordinglyp. However, the current maintains a continuous triangular waveform and does not fall to zero confirming continuous conduction mode (CCM).

- **Switch node analysis**: Compared with the ideal simulation the Switch-node voltage toggles between approximately 0V to 41.41V, representing the MOSFET switching operation between on and off state. A transient overshoot of 49.41V is observed during turn-off corresponding to approx. 7.4V overshoot above input voltage, caused by parasitic inductance/capacitance in the switching loop. The RC snubber damps switching transients and limits excessive ringing resulting in smoother switch node transitions and reduced voltage stress on MOSFET. While the peak switch node voltage remains within the safe operating voltage range of the selected MOSFET.

- **Diode current**: Comparing the diode current with the ideal one buck converter circuit a transient current spike/ringing is observed at turn-on and turn-off due to reverse recovery/parasitic LC interaction in the switching loop.

- **Duty cycle**: The practical duty cycle observed in simulation is higher than the ideal theoretical value predicted by ideal buck converter relation due to non-ideal voltage drops across real components. These include MOSFET conduction loss, diode forward voltage drops, Inductor DCR, Source resistance and switching losses. So as a result, a higher duty cycle is required to maintain the desired output voltage under practical operating condition.

- **Efficiency**: The converter efficiency is calculated by comparing average output power to average input power. Output power is determined from the product of average output voltage and load current while input power is determined from the product of average input voltage and source current. The calculated efficiency reflects the cumulative impact of the conduction losses, parasitic resistance, diode forward drop, and switching losses in the practical converter model.

Calculation:

$\eta(\%) = P_{out} / P_{in}$: ($V_{out(avg)}$ * $I_{out(avg)}$ / $V_{in(avg)}$ * $I_{in(avg)}$) * 100 = ((11.307 * 4.713)/(41.86 * 1.378) * 100 = (53.29/ 57.68)*100 = 92.38%

```
vin_avg: AVG(V(n001) )=41.8621431628 FROM 0.008 TO 0.01
iin_avg: AVG(I(V1) )=-1.37856781064 FROM 0.008 TO 0.01
```

Source current is negative by LTspice sign convention(Positive current flows into the positive terminal): Magnitude is used for the power calculation.
By calculation we can observe the simulated converter efficiency under non-ideal operating conditions is ~ 92.4% demonstrating good practical performance for a 60W asynchronous buck converter. Efficiency degradation from ideal case is attributed to semiconductor conduction losses, switching loss, passive component parasitics and snubber dissipation.

7. Ideal vs Non-ideal performance comparison:

Parameter	Version 1 (ideal)	Version 2 (non ideal)	Deviation	Impact
Vout	11.716 V	11.307V	3.49%	Output reduced
Iout	4.885A	4.71A	3.58%	Output reduced
Vripple	35 mv	48.4 mv	38.28%	Ripple increased
Frequency	199.906KHz	200.14KHz	~0.1%	Gate drive accurate
Δ iL	1.31A	1.30A	~0.76%	Slight deviation
Efficiency	Near ideal	92.4 %	-	Real losses introduced
Switch node	Clean	Overshoot + ringing	-	Parasitic effect visible
Duty cycle	28.95%	30.81%	6.42%	Loss compensation
CCM condition (ILmin > 0)	4.229A	4.06A	3.9%	CCM Confirmed

8. Component stress validation:

Component	Rated	Observed stress	Margin
MOSFET Vds	120V	49.41V	~2.43x
MOSFET Current	8.6A	5.36A	~1.6x
Diode Vr	100V	42V	~2.38x
Diode If	20A	7.07A	~2.83x

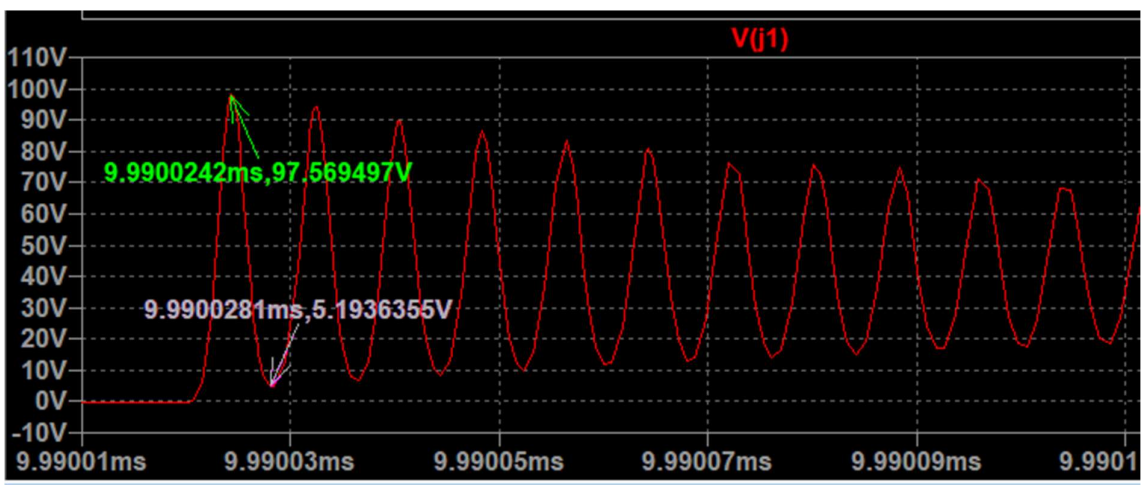
9. Loss breakdown:

Loss component	Formula	Estimate Value
Diode Conduction	Vf*IL(avg)*(1-D)	0.4*4.71*0.692 = 1.30W
MOSFET conduction	IL(avg)^2*Rds*D	4.71^2*0.019*0.308 = 0.13W
Inductor DCR	IL(avg)^2*DCR	4.71^2*0.05 = 1.11W
Source resistance	IL(avg)^2*Rs	4.71^2*0.1 = 2.22W
Snubber	0.5*C*V^2*f	0.5*1n*42^2*200K = 0.176W
Total loss	-	~ 4.936W

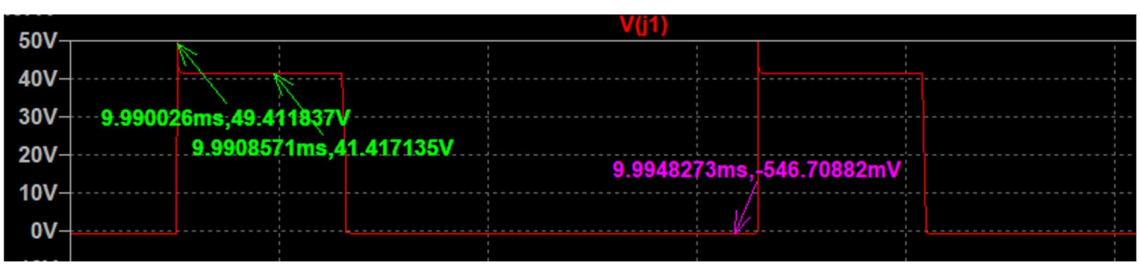
10. Overshoot and ringing effects analysis:

Parameter	Without snubber	With snubber	Impact
Peak switch node Volt.	97.56V	49.41V	49.35% Improved
Overshoot above Vin	56.15V	8V	85.7% reduced
Ringing p-p	92.61V	Negligible/suppressed	Significantly reduced
Ringing condition	Severe underdamped	Damped/Eliminated	Improved

Without snubber simulation:



With snubber simulation:



11. **Snubber effectiveness analysis:** Comparison of switch-node waveform with and without RC snubber confirms significant damping of ringing and reduction in overshoot, thereby lowering voltage stress on the MOSFET and improving switching waveform quality.

12. **Design validation:** The converter successfully performs practical step-down conversion close to the target 12V under non-ideal conditions while maintaining CCM operation, acceptable ripple, controlled switching transients and good practical efficiency.

13. **Key design learnings:**

    - Non-ideal losses increase required duty cycle beyond theoretical prediction.
    - Diode forward drop is dominant efficiency limiter in asynchronous topology.
    - Snubber network effectively mitigates switch-node ringing.
    - Practical parasitics significantly influence ripple and transient response.

14. **Limitation/Future improvements:** Future improvements may include synchronous rectification to replace the Schottky diode for higher efficiency, closed-loop feedback regulation for tighter voltage control and PCB parasitic extraction for hardware-correlated simulation.

15. **Conclusion:** The non-ideal simulation results closely match expected practical behaviour. Compared to the ideal design, output voltage and efficiency reduce due to semiconductor losses and passive parasitics, while output ripple and switching transients increased due to ESR and switching parasitics. The RC snubber effectively suppresses ringing and limits switch-node overshoot. Overall, the converter maintains CCM operation and achieves 92.4% efficiency under realistic operation conditions.